Student ID :	**24CS104**	Student Name:	**Dharmik Vaghani**		
Subject code & Name :	ITUE301: Advanced Web Development Frameworks	Practical :	5	Academic Year :	2026-27

Task - 5

Aim

To integrate MongoDB NoSQL database into the Express application using Mongoose ODM, designing robust data models with schema definitions, data type validations, required constraints, default timestamps, and performing asynchronous database CRUD operations with error catching.

Code

GitHub Link / Repository Link: https://github.com/Dharmikvaghani/AWDF-PRACTICAL.git

Key Implementation Highlights:

1. config/db.js: Mongoose connection management with retry and graceful shutdown.
2. models/Task.js: Strict Mongoose schema with type validation, enum priority constraint, and timestamps.
3. Asynchronous controller refactoring using async/await and Mongoose queries (Task.find, Task.create).

```
// models/Task.js
const taskSchema = new mongoose.Schema({
  title: { type: String, required: [true, 'Task title required'], trim: true },
  description: { type: String, default: '' },
  priority: { type: String, enum: ['low', 'medium', 'high'], default: 'medium' },
  completed: { type: Boolean, default: false },
}, { timestamps: true });
export default mongoose.model('Task', taskSchema);
// config/db.js
export const connectDB = async () => {
  await mongoose.connect(process.env.MONGO_URI);
  console.log('☑ MongoDB Connected Successfully');
};
```

Output

Output

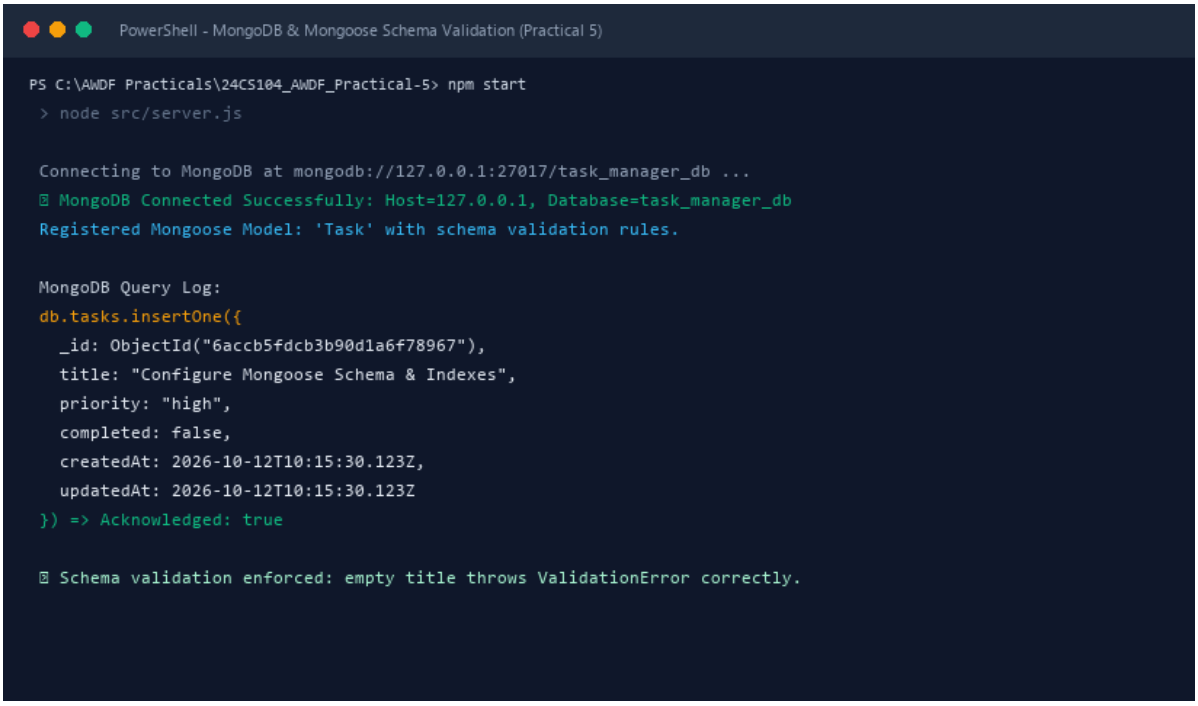


Figure 5: Execution Output & Functional Verification for Practical 5